

# SIMATS ENGINEERING

## ASSESSMENT TOOL 1

|                                  |                                                                                                                                               |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>COURSE NAME</b>               | COMPUTER NETWORKS                                                                                                                             |
| <b>COURSE CODE</b>               | CSA07                                                                                                                                         |
| <b>COURSE OUTCOME COVERED</b>    | CO5: Design network topologies — star, mesh, bus, and hybrid — using networking devices, transmission media, and cabling standards (BL2, BL5) |
| <b>ASSESSMENT TOOL WEIGHTAGE</b> | 40%                                                                                                                                           |
| <b>QUESTIONS</b>                 | 3                                                                                                                                             |
| <b>TOTAL MARKS</b>               | 30                                                                                                                                            |

### Annexure A

### Constraint-Based Problem Solving

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#### Code of Conduct

I (Y.Varun kumar reddy /192424230 ) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Y.Varun kumar reddy

# SIMATS ENGINEERING

## ASSESSMENT TOOL 1 — COMPUTER NETWORKS (CSA07)

CO5: Constraint-Based Problem Solving — Annexure A Answers

### Question 1: Branch Office Connectivity in a Remote Rural Location

A multinational company is opening a branch office in a remote rural area with no wired broadband access. The office needs reliable connectivity for cloud applications, video conferencing, and secure communication with headquarters, on a limited budget.

#### Step 1: Identify the Constraints

- **Bandwidth:** must support cloud apps and video conferencing (moderate-to-high bandwidth, low latency for VC)
- **Availability:** no wired/fibre broadband in the area
- **Budget:** limited capital and operating expenditure
- **Reliability:** business operations must not be interrupted
- **Scalability:** solution should grow with future headcount/traffic
- **Security:** sensitive communication with HQ must stay confidential

#### Step 2: Recommended Internet Connectivity

- **Primary link — 4G/5G LTE Wireless WAN:** cellular broadband routers are quick to deploy, need no cabling, and offer 20–100+ Mbps in most rural areas covered by a tower. This is the most cost-effective 'last-mile' option for remote sites.
- **Backup link — VSAT (satellite broadband):** used only as a failover for locations with poor or no cellular coverage; higher latency and cost, so it is kept as a secondary path, not primary.
- **Dual-WAN failover router:** automatically switches from the primary 4G/5G link to the VSAT/backup link if the primary fails, giving near-uninterrupted connectivity without duplicating full-time satellite cost.

#### Step 3: Networking Devices Required

| Device                              | Purpose                                                              |
|-------------------------------------|----------------------------------------------------------------------|
| 4G/5G LTE + Wi-Fi router (dual-WAN) | Primary internet access and local Wi-Fi/LAN distribution             |
| VSAT modem (backup)                 | Failover connectivity when cellular link is down                     |
| Firewall / UTM appliance            | Perimeter security, intrusion prevention, content filtering          |
| VPN-capable router / concentrator   | Secure site-to-site tunnel to HQ                                     |
| Managed Layer 2/3 switch            | Local wired connectivity for PCs, IP phones, video-conferencing unit |
| Wireless Access Point (optional)    | Wi-Fi coverage for laptops/mobiles in the office                     |

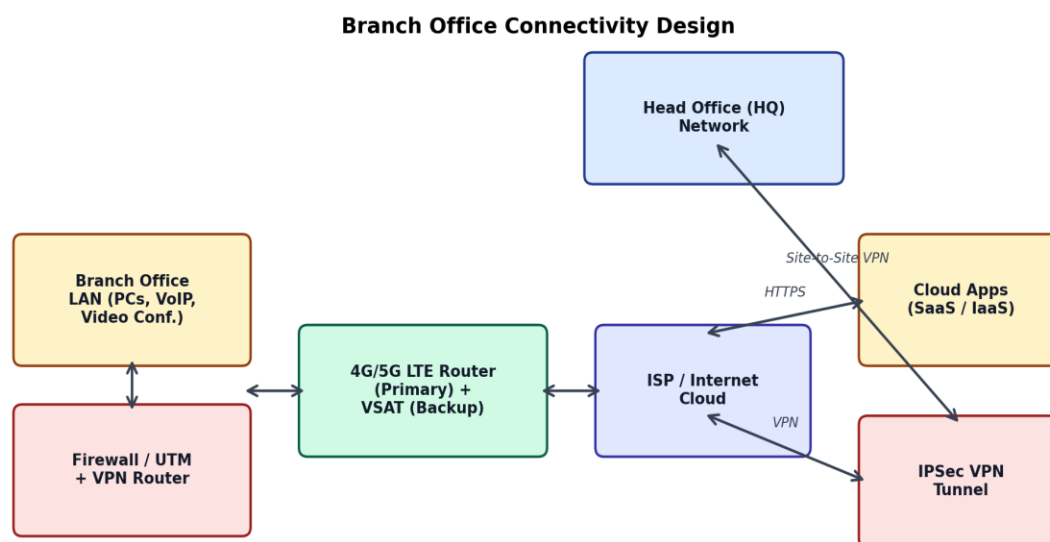


Fig 1.1 — Branch office connectivity design using dual-WAN wireless links and VPN to HQ

#### Step 4: Security Mechanisms

- Site-to-site IPSec VPN tunnel between branch router and HQ firewall for encrypted communication.
- Next-generation firewall (NGFW) with stateful inspection and intrusion prevention (IPS).
- WPA2/WPA3-Enterprise for local Wi-Fi with 802.1X authentication.
- Cloud application access secured with HTTPS/TLS and multi-factor authentication (MFA).
- Regular firmware updates and centralized remote monitoring from HQ NOC.

#### Step 5: Justification

- **Bandwidth:** 4G/5G provides sufficient throughput for cloud apps and video conferencing without laying expensive fibre
- **Reliability:** dual-WAN (cellular + satellite backup) removes single point of failure, ensuring uninterrupted operations
- **Cost:** wireless WAN avoids high fibre trenching cost in a rural area; VSAT is used only as standby, keeping OPEX low
- **Scalability:** additional bandwidth can be purchased from the telecom operator as the branch grows, and extra APs/switches can be added without re-architecting the network

## Question 2: Campus-Wide Wireless Network for 3,000+ Students

A university needs to provide Wi-Fi across classrooms, laboratories, libraries, and hostels for over 3,000 students, using a limited number of access points (APs), while minimizing interference and keeping access secure.

#### Step 1: Identify the Constraints

- **User density:** 3,000+ simultaneous users across multiple buildings
- **Coverage:** classrooms, labs, library, hostels — different building types and wall densities
- **Budget:** limited number of APs can be purchased/deployed
- **Interference:** overlapping Wi-Fi cells and neighbouring APs must not degrade performance

- **Security:** only authorized students/staff should get network access

## Step 2: Wireless Network Design

- **AP placement:** conduct a Wi-Fi site survey; place APs at ceiling height in high-density areas (classrooms, library reading halls) with a coverage radius of ~20–30 m indoors, and stagger AP positions on alternating floors to reduce co-channel interference
- **Capacity planning:** budget roughly 25–40 concurrent users per AP for Wi-Fi 6 (802.11ax) APs, which support OFDMA and MU-MIMO for far higher effective user density than older standards — this reduces the total AP count needed for the same user load
- **Centralized management:** a Wireless LAN Controller (WLC) manages all APs centrally for consistent SSIDs, load balancing, band-steering, and automatic channel/power adjustment (RRM)

## Step 3: Frequency Band Selection

| Band                               | Use Case                                                                                                                                             |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2.4 GHz                            | Longer range/wall penetration for hostels and open areas; limited to 3 non-overlapping channels (1, 6, 11) — used sparingly to control interference  |
| 5 GHz                              | Primary band for classrooms, labs, and library — more non-overlapping channels, higher throughput, shorter range which naturally limits cell overlap |
| 6 GHz (Wi-Fi 6E, if budget allows) | Additional clean spectrum for high-density zones such as auditoriums and libraries                                                                   |

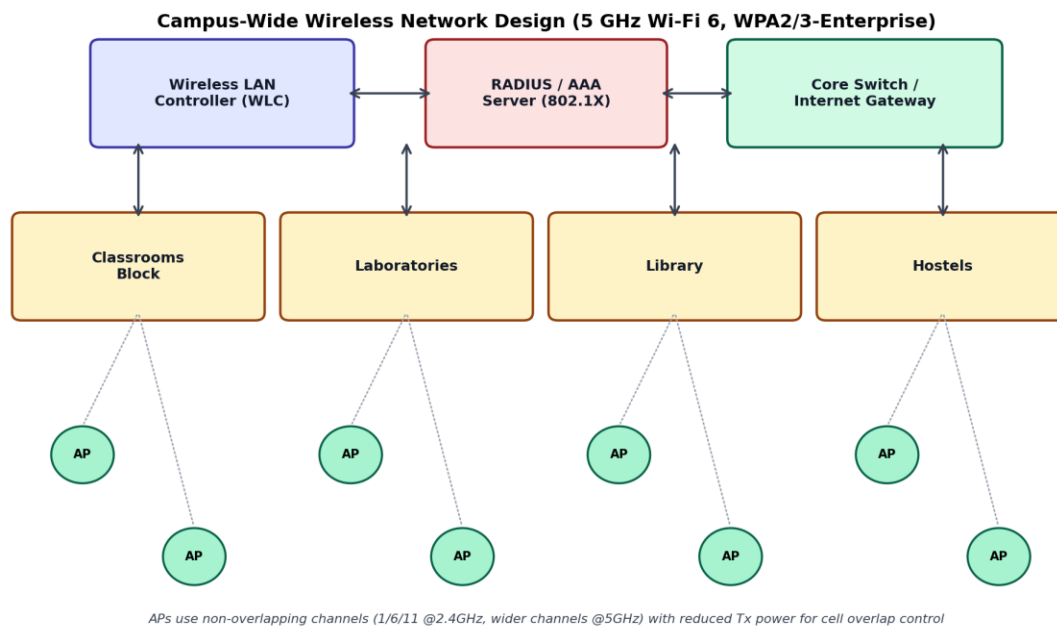


Fig 2.1 — Campus wireless design: centralized WLC/RADIUS with AP clusters per building

## Step 4: Authentication Methods

- WPA2/WPA3-Enterprise with 802.1X authentication against a RADIUS server tied to the university's directory (student/staff ID).
- Separate SSIDs: one for students/staff (802.1X/certificate-based) and one guest SSID (captive portal, rate-limited, isolated VLAN).

- Role-based VLANs to segment student, faculty, and administrative traffic.

### Step 5: Justification — Balancing Coverage, Performance, Security, and Cost

- **Coverage:** site-survey-based AP placement with 5 GHz as primary band ensures reliable signal in every building without excess AP purchases
- **Performance:** Wi-Fi 6 APs with MU-MIMO/OFDMA raise per-AP user capacity, letting a limited AP budget still serve 3,000+ students
- **Security:** 802.1X + RADIUS + WPA2/3-Enterprise ensures only authenticated users connect, while guest isolation protects the internal network
- **Cost:** centralized WLC and RRM-based channel/power planning avoid over-provisioning APs, keeping capital cost within budget while meeting coverage needs

## Question 3: Secure Network Architecture for a Financial Institution

A financial institution must upgrade internal network security to protect sensitive customer data and online banking services, while complying with strict regulations and keeping performance and cost within acceptable limits.

### Step 1: Identify the Constraints

- **Security:** protect sensitive customer and transaction data from external/internal threats
- **Compliance:** meet regulatory standards (e.g., PCI-DSS, RBI/ISO 27001-type guidelines) for financial data handling
- **Performance:** security controls must not introduce unacceptable latency for banking transactions
- **Budget:** solution must be achievable without large recurring operational cost

### Step 2: Network Segmentation Strategy

- **DMZ (Demilitarized Zone):** public-facing online banking web/application servers are placed here, isolated from the internal network
- **Core banking VLAN:** highest-security segment holding core banking servers and databases, accessible only to authorized internal systems
- **Employee workstation VLAN:** separates staff PCs from servers to limit lateral movement if a workstation is compromised
- **Customer data / database VLAN:** isolated segment with strict access control lists (ACLs) for compliance
- **Guest/third-party VLAN:** isolated network for vendors/auditors with no route to internal banking segments

### Step 3: Firewall Deployment

- External Next-Generation Firewall (NGFW) between the internet and the DMZ — performs deep packet inspection, application-layer filtering, and threat intelligence-based blocking.
- Internal firewall/segmentation layer between the DMZ/core network and internal VLANs, enforcing least-privilege rules between segments.
- Firewall rules follow the principle of least privilege — only explicitly required ports/protocols are allowed between segments.

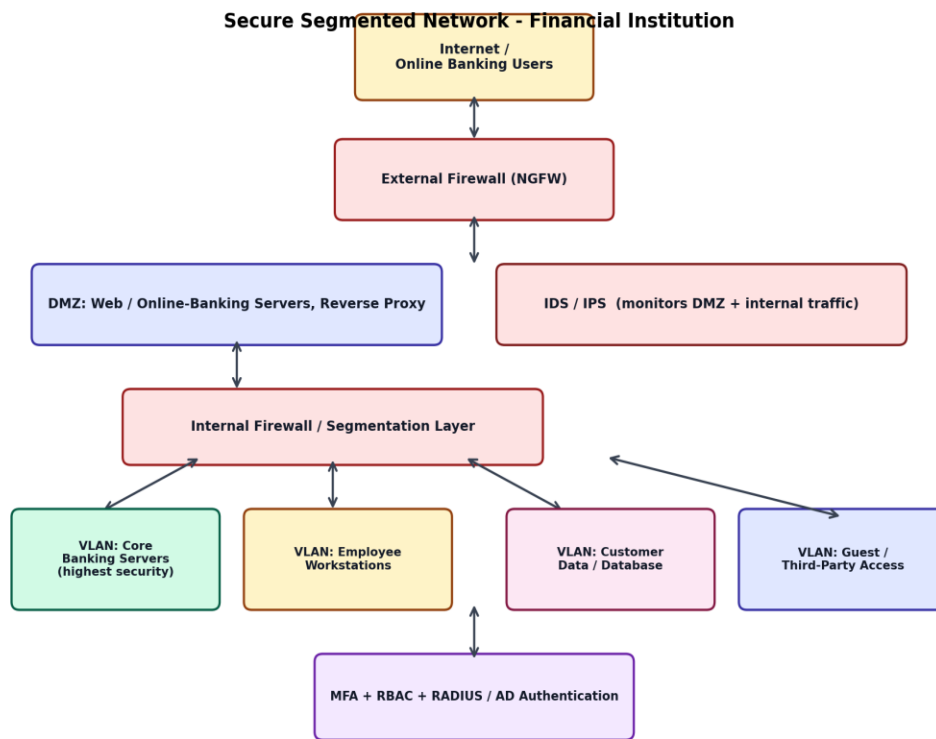


Fig 3.1 — Segmented secure architecture: DMZ, internal firewall, VLAN segmentation, IDS/IPS and authentication layer

#### Step 4: Authentication and Intrusion Detection

- **Authentication:** Multi-Factor Authentication (MFA) for all employee and customer-facing logins, combined with Role-Based Access Control (RBAC) and centralized RADIUS/Active Directory authentication
- **Intrusion Detection/Prevention (IDS/IPS):** deployed at the DMZ boundary and on internal segment links to continuously monitor traffic, detect anomalies, and automatically block malicious activity
- **Logging & monitoring:** centralized SIEM (Security Information and Event Management) collects logs from firewalls, IDS/IPS, and servers for compliance audit trails

#### Step 5: Justification

- **Security:** layered defense — perimeter NGFW, DMZ isolation, internal segmentation, and IDS/IPS — ensures no single control failure exposes core banking data
- **Compliance:** VLAN-based segmentation with strict ACLs and MFA/RBAC directly satisfies data-isolation and access-control requirements of financial regulations
- **Performance:** segmentation and dedicated firewall appliances (rather than software-only filtering) keep inspection overhead low, preserving transaction speed
- **Budget:** reuse of existing switches with VLAN tagging (802.1Q) plus a two-tier firewall design avoids the cost of a fully rebuilt network, while still meeting security goals

— End of Answers —